

Theorem Solver

Given a math statement & int n is there a proof in ZF set theory of the statement with at most n symbols.

Computable but may take exponential time.

Given a solution, possible to verify its validity in poly time.

Undecidable if we take away n symbol constraint.

Consider it a finite version of the halting problem.

Allows to solve **practically** all open problems.

→ Do we care if the proof is exponentially long.

Nondeterministic Polynomial Time (NP)

NP is the class of languages decidable in polynomial time by a single tape NTM.

Def 1 $NP = \bigcup_k NTIME(n^k)$

Def 2

Class of all languages for which there exists a poly time DTM M & a polynomial $p(n)$, s.t., for all $x \in \{0,1\}^n$:

$x \in L \iff$ there exists $y \in \{0,1\}^{p(n)}$ s.t. $M(x,y)$ accepts.

Think of y as the candidate solution that can be verified in polytime by DTM.

Why are the two definitions equivalent?

Have the NTM guess y & check whether it works.

NP-Hard & NP-Complete

NP-Hard contains problems that are **at least** as hard as any NP problem.

→ doesn't have to be in NP.

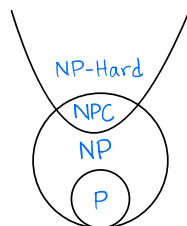
Def L is NP-Hard if given an oracle for L , we can solve any NP problem in deterministic polytime.

L is NP-Hard if $NP \leq P^L$.

Polytime DTM with access to oracle for L can solve any NP problem

NP-Complete (NPC)

Problems that are both NP-Hard and in NP.



World-view if $P \neq NP$

The Doh Problem

Given a poly time DTM M , $\overbrace{\text{decision versus search}}^{\text{version of problem}}$ $\underbrace{\text{does there exist/find}}_{\text{polynomially related to each other}}$ input y that causes $M(y)$ to accept.

Doh is in NP

Given solution y , check if $M(y)$ accepts.

Doh is in NP-Hard

Suppose have access to oracle L for Doh. Given any NP problem, by definition, there must exist a poly time DTM $M'(x,y)$ for verifying solution y . Ask oracle for L , whether there exists a y that causes M' to accept after hardwiring x in description of M' .